

Modern cartography & 3D map



2D

(like



Radio



3D

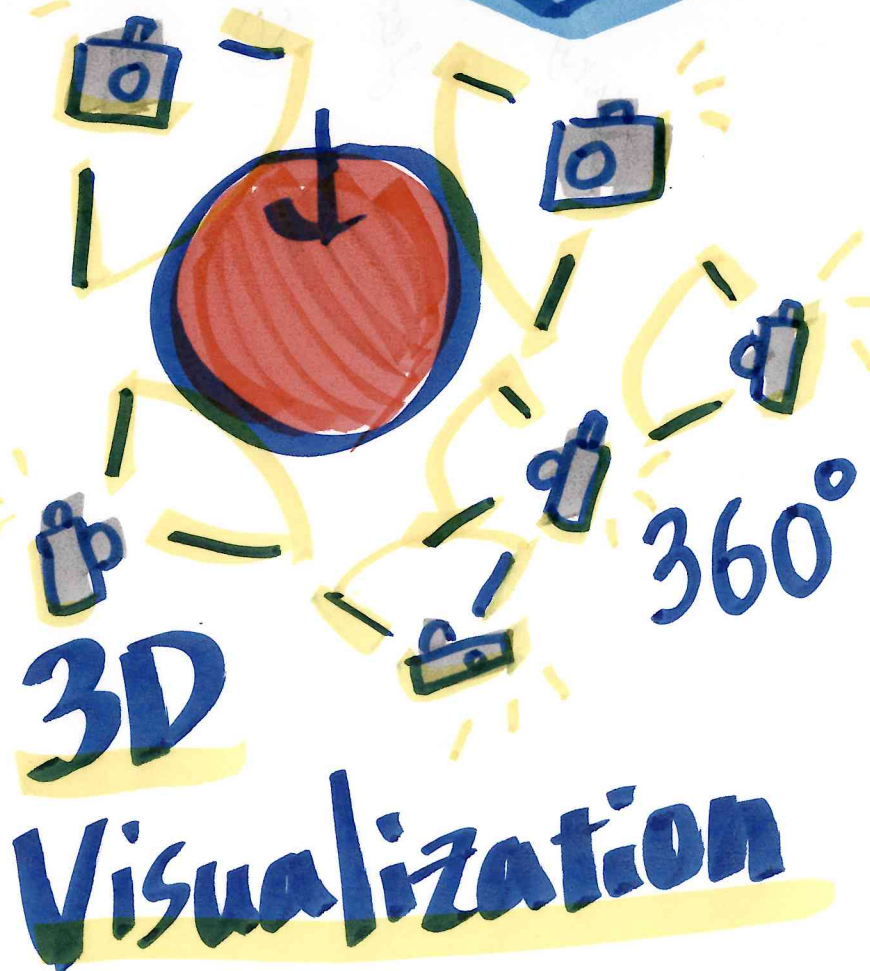
TV!

NEED
INDIVIDUAL
DATA!



Rachel Hwang
engineer & artist

| A115050 大浦万里愛



3D
Visualization

- ⊕ geospatial data
- ⊕ in common
- ⊕ widely used
- ⊕ cheap
- ⊕ easy



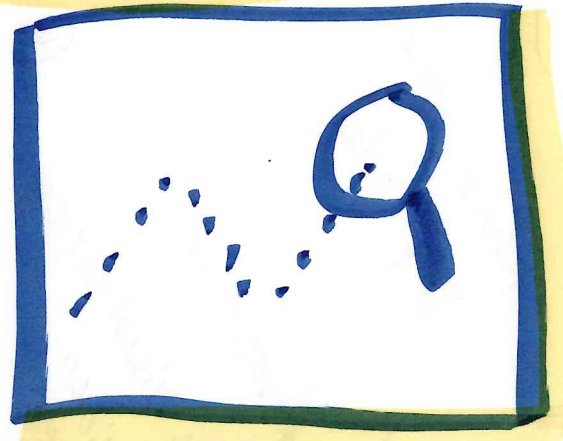
OPEN SOURCE
DATA



Visualization
TOUR
in Your
HAND!

WHY & WHERE are shown on MAP

MAPS



Paper ...

Digital

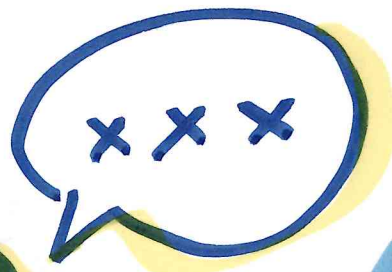
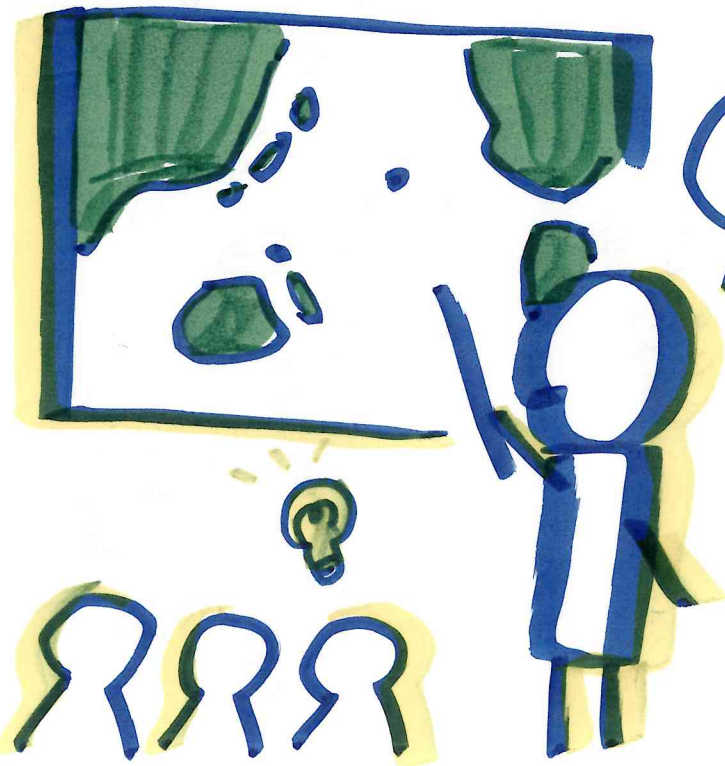
Geo- is the
Keyword in
next FUTURE



Joseph.K
GEOGRAPHER

Worth
1000s of Pics

GPS 1A115050大浦万野
GIS (地理情報システム)



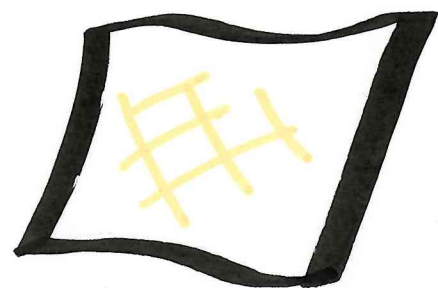
ooo



Whys of Where

1A117146 古山 結衣

① 地図の進化



★紙の場合

情報を足せない...

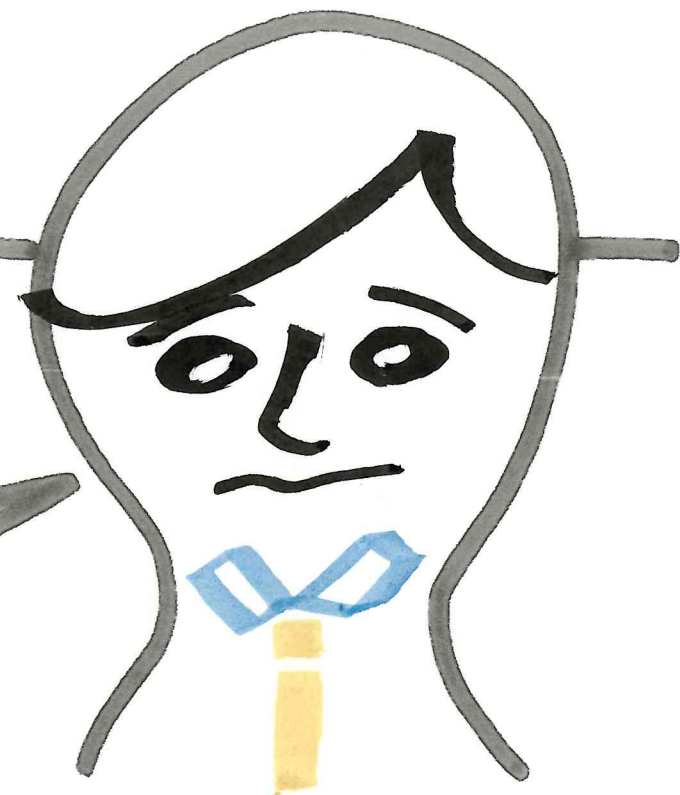
大きさも変えられない...

持っていてのもなんか...

良い地図とは
ユーザーに良い問を与えられるコト。

GIS

使えば何でも??
わかるよ!!



Joseph
Kerski

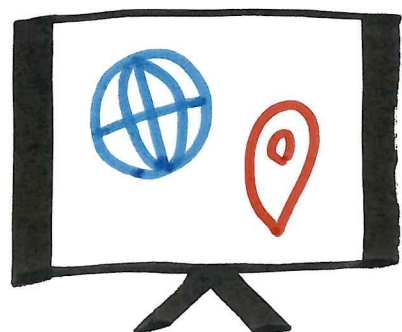
災害時の情報
世界規模で!

人口もわかる、年齢も
どの授業でも地図は
使われている! うバき!

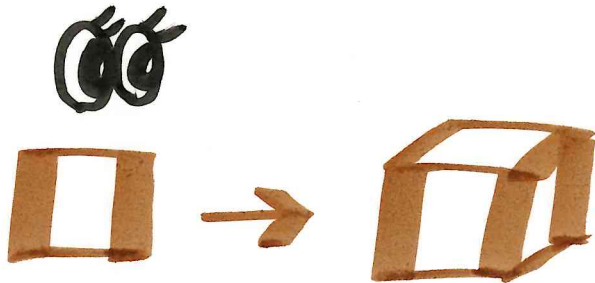
② 今の技術なら! GIS



or



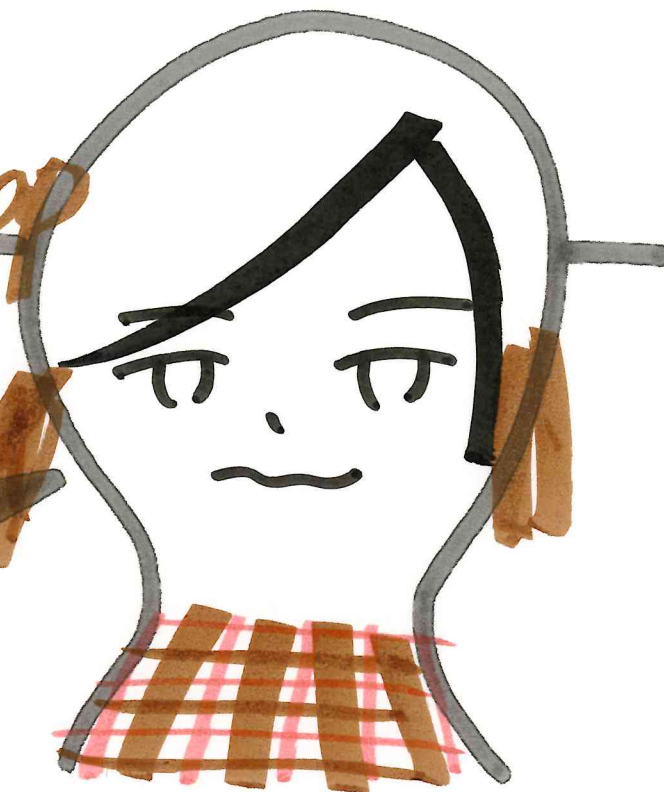
Modern cartography and the 3D map

2D → 3D 

3Dの地図において

⑥ 求めているのは

1A117146 古山結衣
3Dの世界に住んで
いるから3Dの
情報が大事でショ。



Rachel Hwang

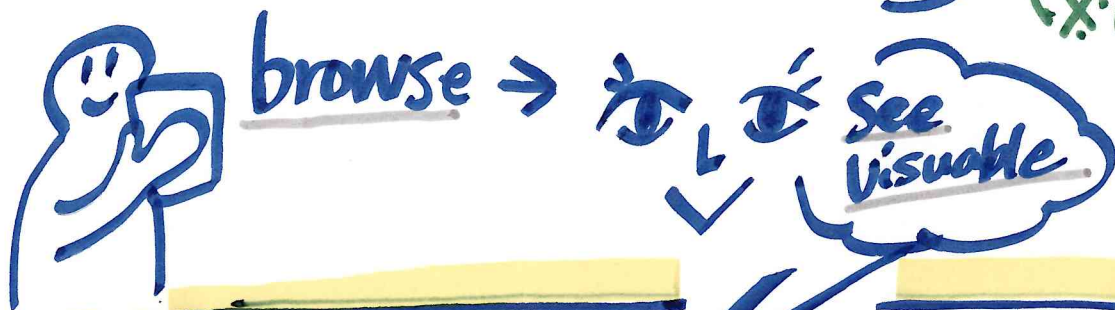
natural environment

日常に根ざしたものの雨情報やレストランなどのReal.
日のあたり方とか。インフラ情報

発展なども もっと知られるべきじゃないの??

専門家 → 3Dは一般向けであるべき

Modern ~~Mobile~~ Cartography & 3D map Revo!



↳ Digire map

Cartography

3D maps/Geospatial
for
EVERYONE!



Rachel Hwang
engineer & artist

(3D map)

- 3D -
- geospatial (i)
- know ↑ (i)
- map data amount ↑

High Resolution!!



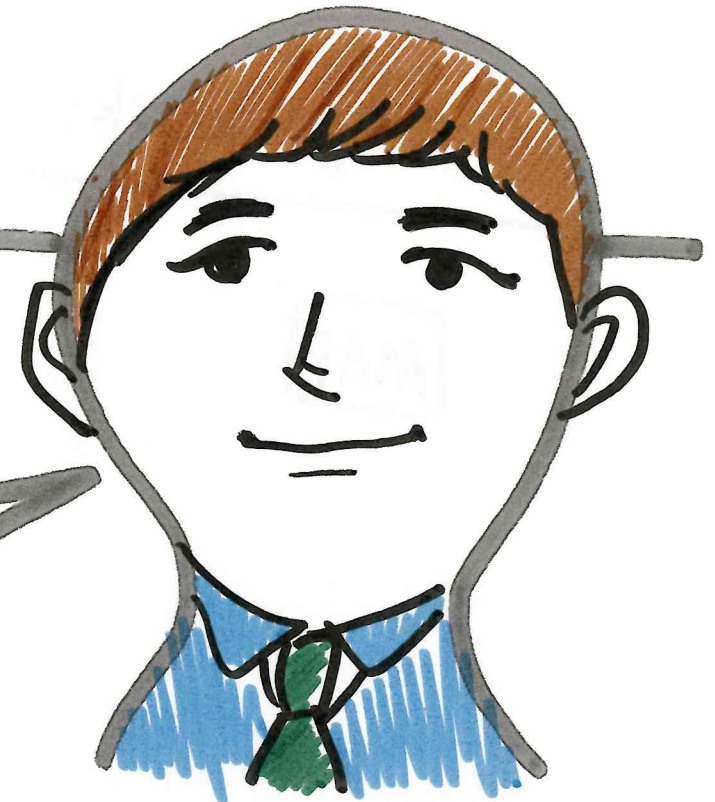
1A116182 原@桜子

Whys Of Where



A good MAPs helps
to a better Q.!

 or 

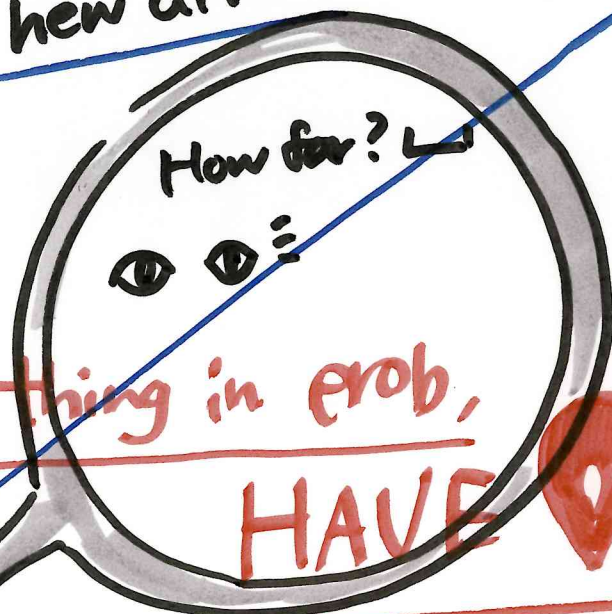


Joseph Kersi

in TEDx TALK

add new info...

teach, bring
new attention



★ Bio, nano...

GIS!

connect  world wide

mixed up w/ , , 

p b l

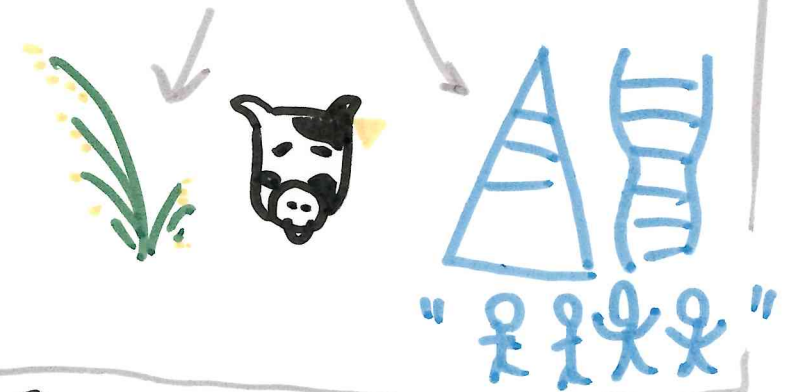
(prob base learning)



Ex

Are GIS...

Remote sensing...



1A116182 原田桜子

"3D MAP" Revolution

IA117052 金田佳奈



Rachel Huang

1 huge step

Sharing Map on the web



2010. 2D -> 3D

WHY we live in 3D

3D Mapping

(Ex) 360° Map
historic data
of earthquake

Geospatial Data

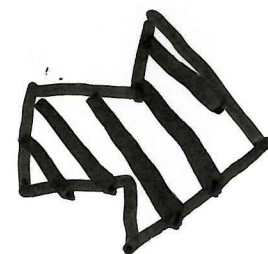
(ex): Urban
Environment

Important
Technology

• lidar
• Photography

↳ widely & cheaply
available

3D MAP
& Technology
For
EVERYONE



Access Availability

(Ex) detail Data for shopping
New Apartment

technology
Application

Difficult

Whys of Where

Maps are

IA117052 金田佳奈



Maps — Thousands of Pic. and Info

 Paper Map.

✗ Not updated. limited.

Spatial Thinking
Mapping Technology

 digital Map.

○ real time info.

 direction. Aspect etc..

Geo tech.
Web Mapping

GLS = Allow us
Many application.

Problem have location.

↓
Why's of Where

good Map teaches  population. Risk of disaster

us to have question

The 3D Map Revolution

なぜ3Dが大事か?

2D $\Rightarrow$ 3D

3D Maps
and
visualizing
geospatial
data.

Rachel Hwang

光を用いたリモートセンシング

主な技術

1. lidar

2. photogrammetry

geospatial
data

高度データ

$\rightarrow$ DSM

Good environment
and
infrastructure

IA117043

小山内 裕博

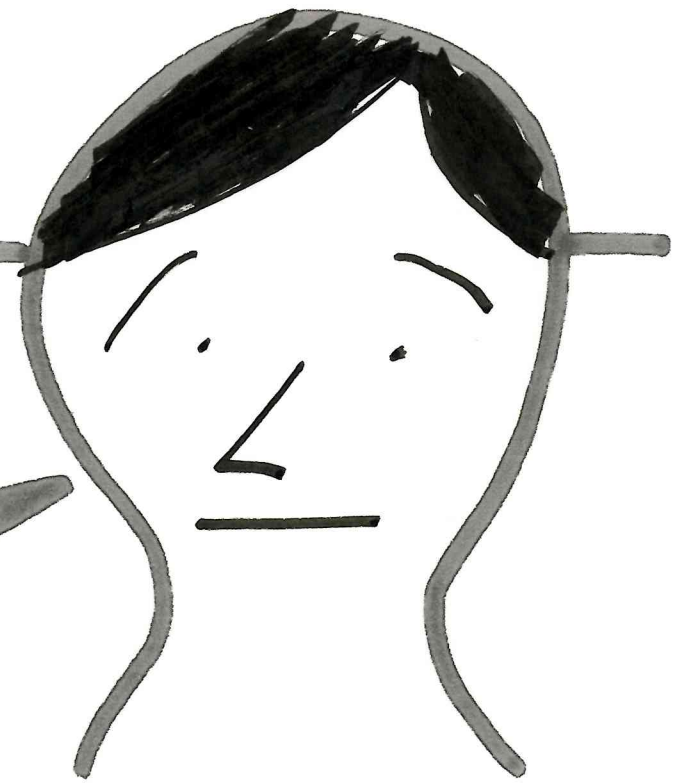
画像から対象の幾何学的特徴を抽出

Whys of where

Paper maps

制限^いがある!

GISで
世界は明るく!



Joseph Kerski

Geo graphic information systems

Arc GIS など様々なデータを
地図上に集約!!

様々な分野の垣根を
超えて、GISは
活用可能!!

illness

population

earthquakes

1A117043

小山内 裕夢

3D Map Revolution

map

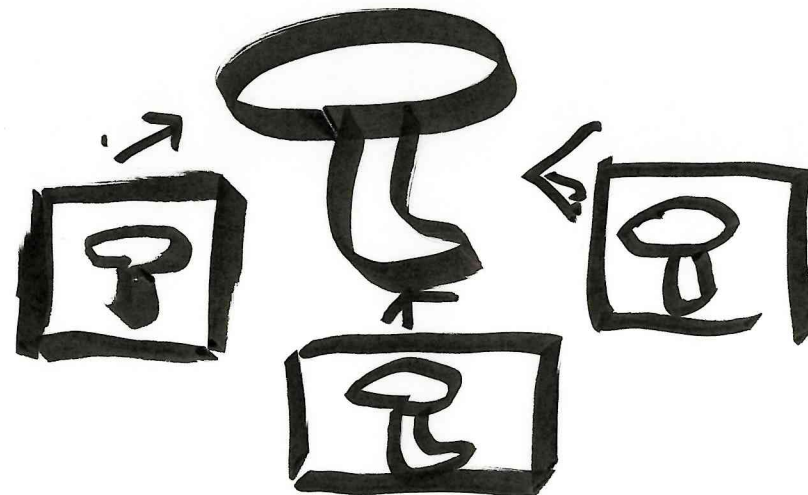
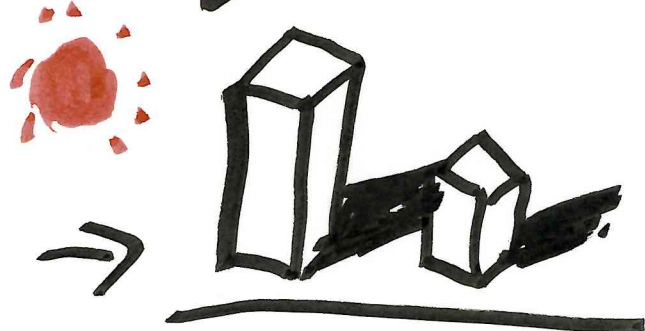
2D → 3D
nowadays tec.

3D Map
need and have
information



Rachel Huang

3D → gives more
information



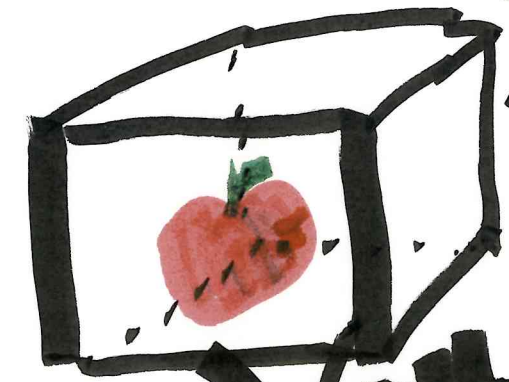
every
one can
help to
create
3D Map

3D model need
many of pic.

You can
see inside

none / none

Lidar

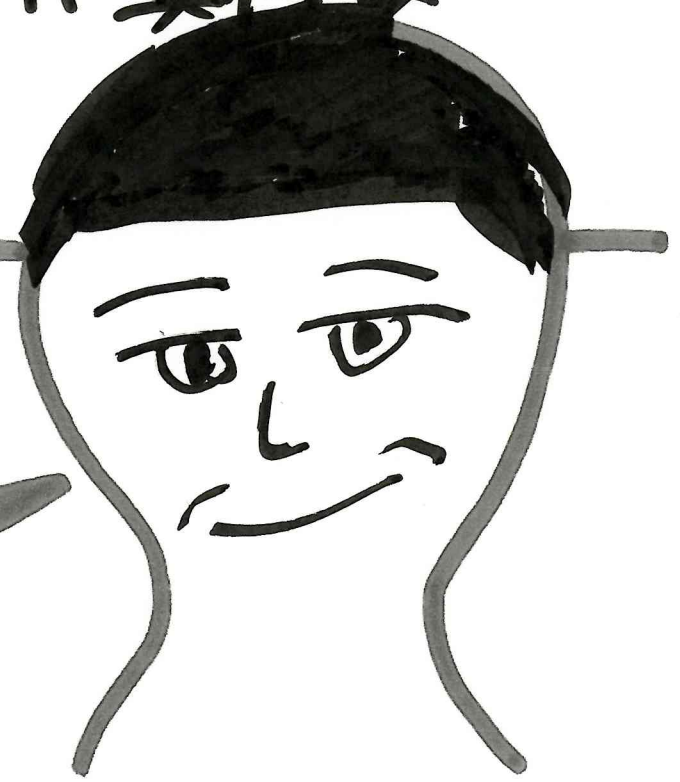


7 Whys of Where.

Paper Maps
can't up date

Limited information

map will
solve global
issue!!



Joseph Kerski

↓ but

digital maps can
No limit data

up date data



GPS, GIS, Pic, data, information

map

what
going

easy to see or.



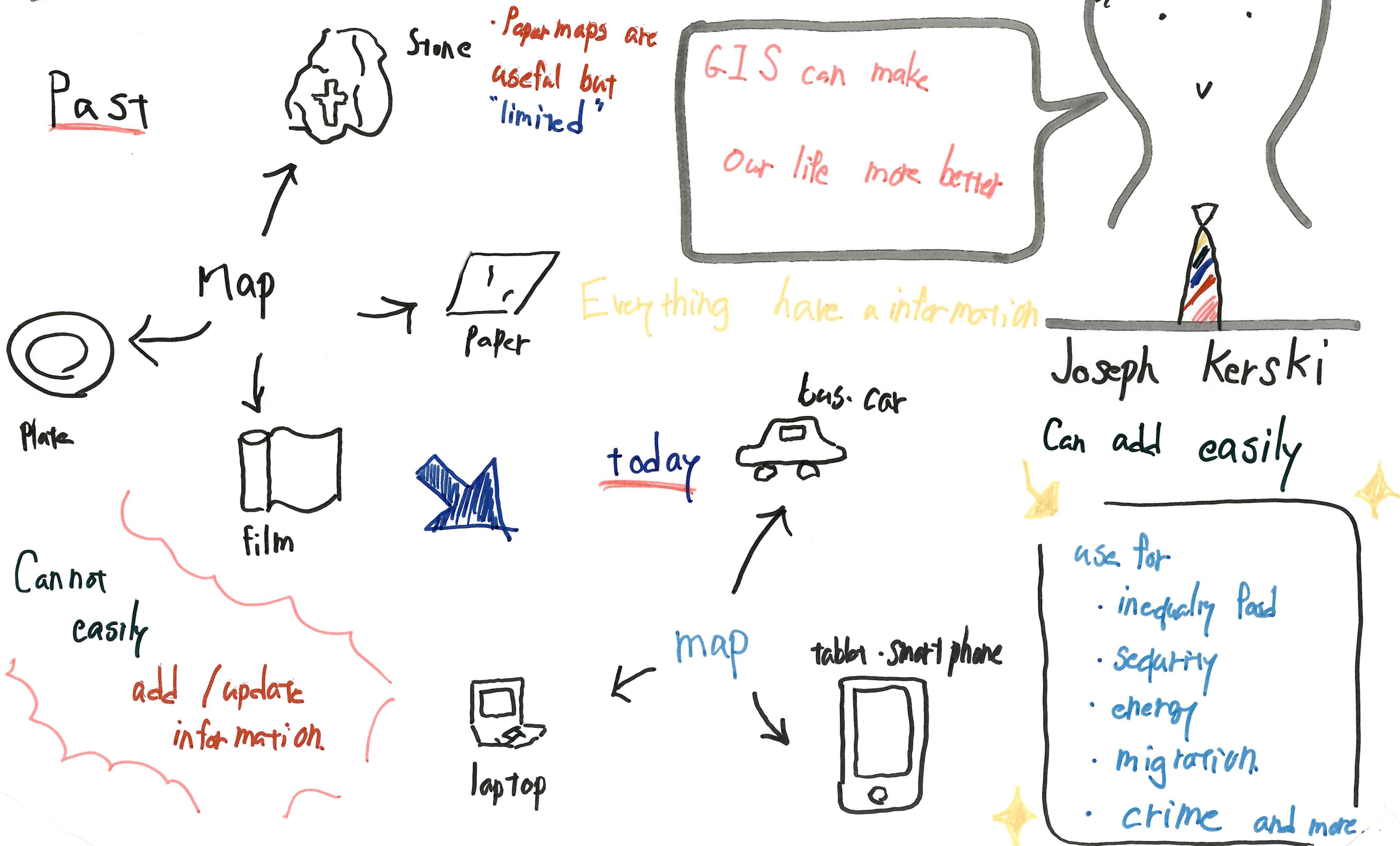
map can
connect all
information

10 data
base

Why's of Where.

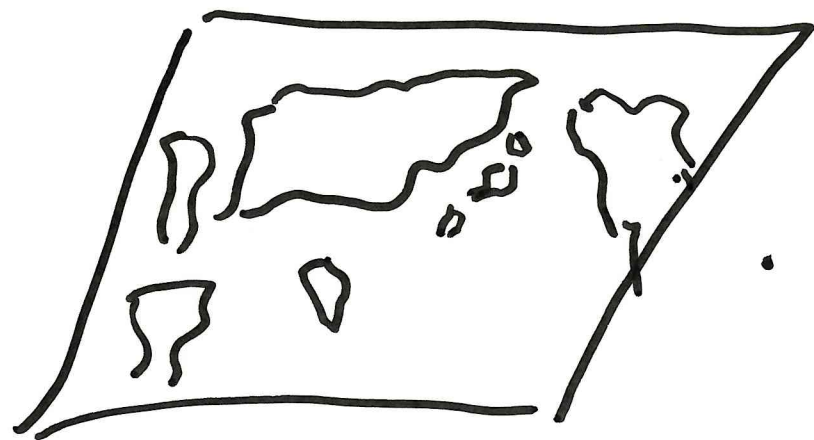
1A117153

松本航



3D Map Revolution

1A117153
松本 航



Paper maps

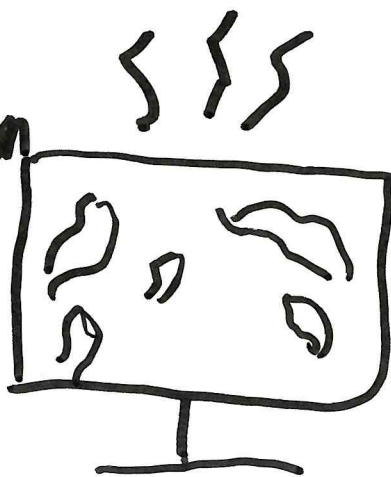
3D Map problems for
every one, not just
experts.



Rachel

1st Step

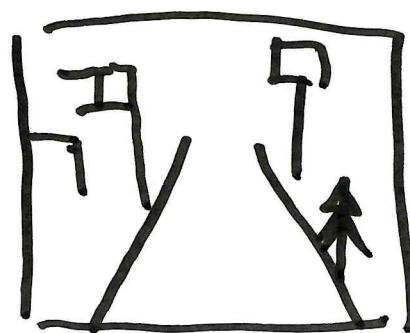
- add information
easily



web map

Visualization
||
Comfortable for using

- everyone could
access map



Street map

- Step that 2D to 3D

• unlocked new dimension

OSM

- Anyone can contribute to Map
- Data are not enough,
so we should take part
in this project !!!

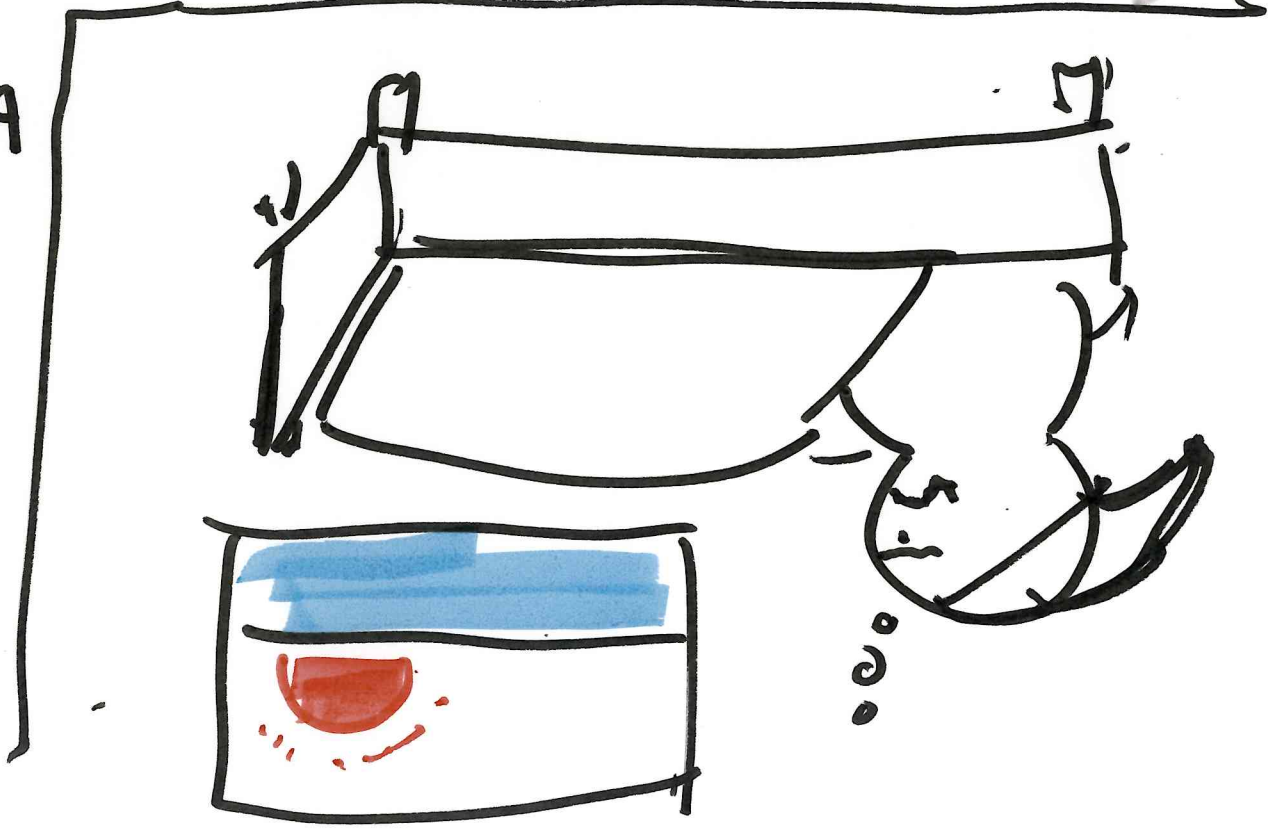
3D Geospatial for Everyone

Work with US!

Can I find a place that I can see a sunset ~~from~~ in a bed?

difficult with 2D

but

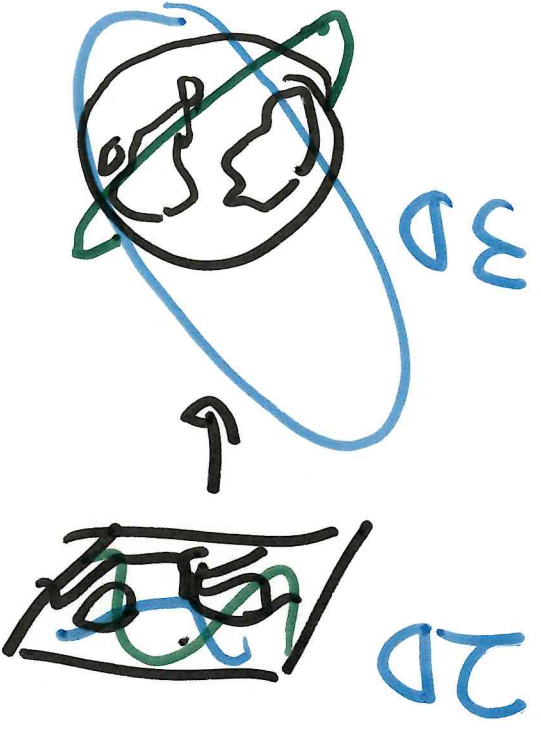
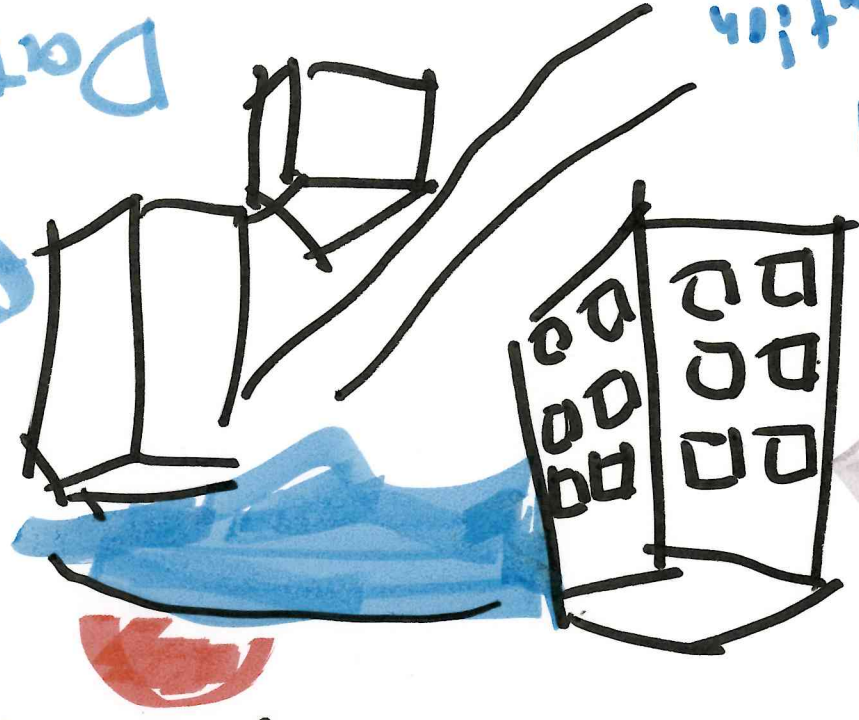


3D Map is possible!

- o Satellite
- o Lidar
- o Photogrammetry

Data

Open data
Standard
Communication



3D

2D

12/17/2022

Rachel Huang

Whys of Where

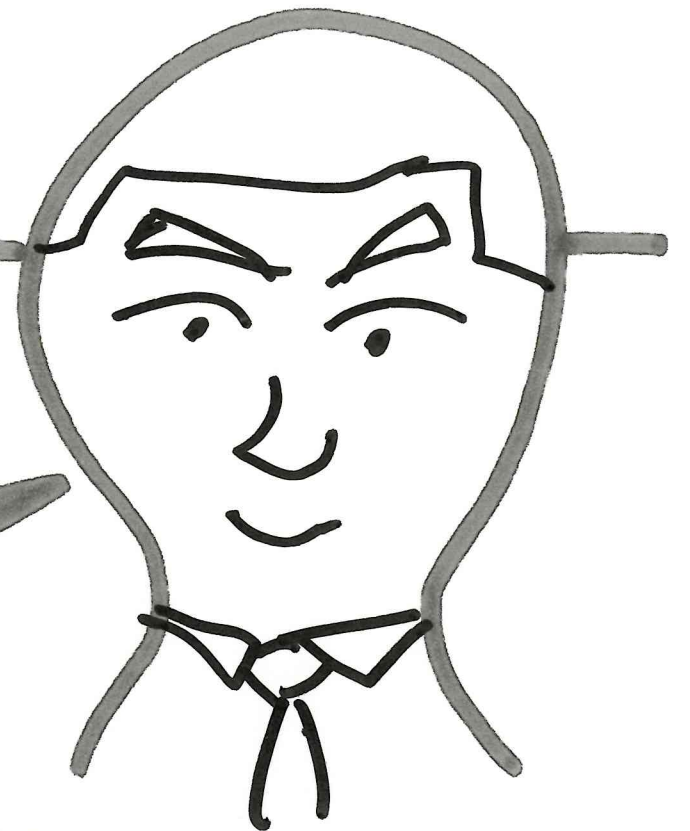
Deep spatial thinking



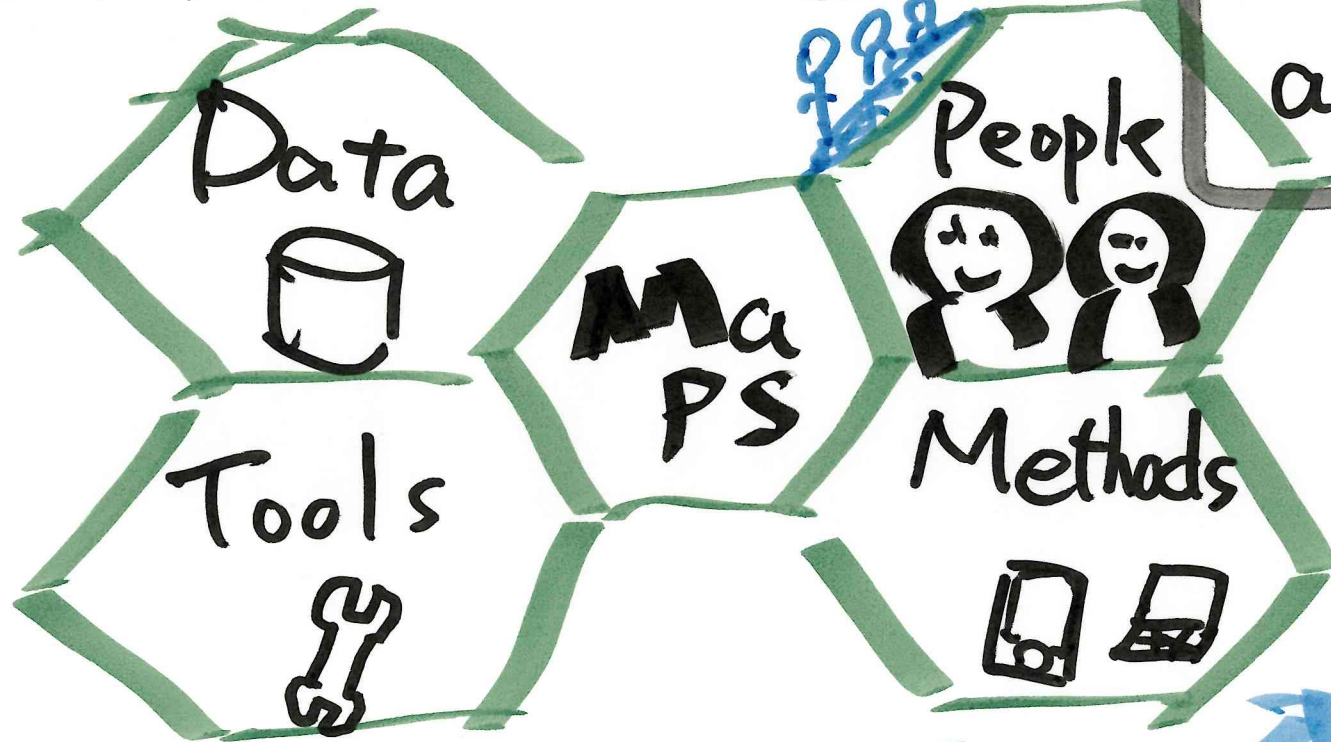
Geo data

Everybody
Can use!

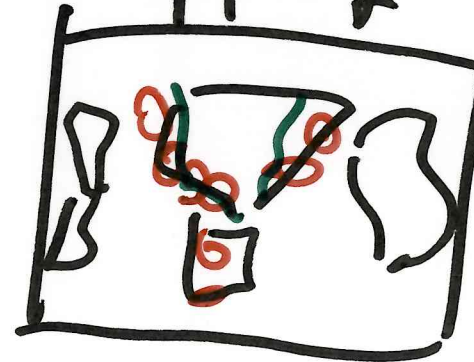
A good Map
teaches you
a good Question!



Joseph Kerski



Mapping



web

Top 10 Issues

Paper map is....

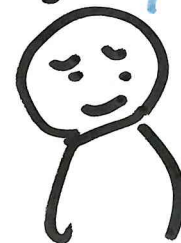


difficult to

- update
- expand....

- realtime mix data
- 3D
- direction
- scalable

why?



you can
see good
question!

All includes
Location

- ☐ climate
- ☐ economic
- ☐
- ☐
- ☐

何しろ